

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

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1. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Given:

Initial Weight (W) = 0.5

Learning Rate (η) = 0.1

Input (X) = 2

Actual Output (Y) = 0.6

Target Output (T) = 1

Step 1: Calculate the Error

$$\text{Error} = \text{Target Output} - \text{Actual Output}$$

$$\text{Error} = 1 - 0.6$$

$$\text{Error} = 0.4$$

Step 2: Calculate the Weight Update

According to the Delta Learning Rule:

$$\Delta W = \eta \times \text{Error} \times \text{Input}$$

$$\Delta W = 0.1 \times 0.4 \times 2$$

$$\Delta W = 0.08$$

Therefore, the weight update is 0.08.

Step 3: Calculate the New Weight

$$\text{New Weight} = \text{Old Weight} + \text{Weight Update}$$

$$\text{New Weight} = 0.5 + 0.08$$

$$\text{New Weight} = 0.58$$

Final Answer: Error = 0.4

Weight Update = 0.08

New Updated Weight = 0.58

Conclusion:

The delta learning rule modifies the weight based on the error between the target and actual output. Since the target output is greater than the actual output, the weight increases from 0.5 to 0.58. This adjustment helps the neuron produce an output closer to the target value.

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is -0.6 . For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected.

Given:

Income (X_1) = 1.5

Credit Score (X_2) = 2

Weight for Income (W_1) = 0.7

Weight for Credit Score (W_2) = 0.5

Bias (b) = -0.6

Step 1: Calculate the Net Input

$$\text{Net Input} = (W_1 \times X_1) + (W_2 \times X_2) + b$$

$$\text{Net Input} = (0.7 \times 1.5) + (0.5 \times 2) - 0.6$$

$$\text{Net Input} = 1.05 + 1 - 0.6$$

$$\text{Net Input} = 1.45$$

Step 2: Apply the Step Activation Function

The step activation function is:

If Net Input ≥ 0 , Output = 1

If Net Input < 0 , Output = 0

Since the calculated net input is 1.45: $1.45 \geq 0$

Final Answer: Net Input = 1.45

Output = 1

Decision = Loan Approved

Conclusion:

The perceptron produces an output of 1 because the net input is positive. Therefore, according to the given perceptron model, the applicant's loan is approved.

9. A Genetic Algorithm is used to maximize the function $f(x) = 2x + 3$ for an initial population of values $\{1, 3, 5, 7\}$. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence.

Given:

Fitness Function: $f(x) = 2x + 3$

Initial Population: $\{1, 3, 5, 7\}$

Step 1: Calculate the Fitness Values

For $x = 1$:

$$f(1) = 2(1) + 3 = 5$$

For $x = 3$:

$$f(3) = 2(3) + 3 = 9$$

For $x = 5$:

$$f(5) = 2(5) + 3 = 13$$

For $x = 7$:

$$f(7) = 2(7) + 3 = 17$$

Fitness Table:

Individual	Fitness
1	5
3	9
5	13
7	17

Step 2: Select the Top Two Candidates

Since the objective is to maximize the fitness function, the individuals with the highest fitness values are selected.

- First candidate: 7, Fitness = 17
- Second candidate: 5, Fitness = 13

Therefore, the selected parents are: 5 and 7

Step 3: Perform Single-Point Crossover

Convert the selected parents into binary form:

$$5 = 0101$$

$$7 = 0111$$

Select a crossover point after the second bit:

$$5 = 01 \mid 01$$

$$7 = 01 \mid 11$$

Exchange the portions after the crossover point.

$$\text{Offspring 1} = 01 \mid 11 = 0111 = 7$$

$$\text{Offspring 2} = 01 \mid 01 = 0101 = 5$$

Therefore, the generated offspring are: 7 and 5

Note: The offspring may be different if a different crossover point is selected.

Step 4: Role of Mutation

Mutation is a genetic operation in which one or more bits of a chromosome are randomly changed.

Mutation helps the Genetic Algorithm by:

1. Maintaining diversity in the population.
2. Introducing new possible solutions.
3. Preventing premature convergence.
4. Helping the algorithm escape local optima.
5. Exploring different areas of the search space.

Conclusion:

The fitness values of the population are 5, 9, 13, and 17. The best two candidates are 5 and 7. Crossover generates new offspring by exchanging portions of the selected parents. Mutation maintains diversity and helps the algorithm avoid getting trapped in a local optimum.

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration.

Given:

Initial Weight (W) = 0.6
Learning Rate (η) = 0.15
Input (X) = 2.5
Actual Output (Y) = 0.7
Target Output (T) = 1

Step 1: Calculate the Error

$$\text{Error} = \text{Target Output} - \text{Actual Output}$$

$$\text{Error} = 1 - 0.7$$

$$\text{Error} = 0.3$$

Step 2: Calculate the Weight Update

Using the Delta Learning Rule:

$$\Delta W = \eta \times \text{Error} \times \text{Input}$$

$$\Delta W = 0.15 \times 0.3 \times 2.5$$

$$\Delta W = 0.1125$$

Therefore, the weight update is 0.1125.

Step 3: Calculate the New Weight

$$\text{New Weight} = \text{Old Weight} + \text{Weight Update}$$

$$\text{New Weight} = 0.6 + 0.1125$$

$$\text{New Weight} = 0.7125$$

Final Answer: Error = 0.3

Weight Update = 0.1125

New Updated Weight = 0.7125

Conclusion:

The actual output is 0.7, while the target output is 1. Therefore, the error is positive and the weight is increased from 0.6 to 0.7125. This weight adjustment helps the neuron reduce the error during training.

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation.

Given:

Training Accuracy increases from 65% to 97%.

Validation Accuracy increases from 63% to 75% and then fluctuates around 72%.

Step 1: Identify the Issue

The model is experiencing overfitting.

- The training accuracy continuously increases and reaches 97%. However, the validation accuracy improves only up to 75% and then decreases or fluctuates around 72%.
- This indicates that the model performs very well on the training data but does not perform equally well on unseen data.

Step 2: Reason for Overfitting

Overfitting occurs when a neural network learns the training data too closely. It may learn unnecessary patterns and noise instead of learning general patterns that work well on new data.

Some common causes of overfitting are:

1. Very complex neural network.
2. Too many parameters.
3. Insufficient training data.
4. Training the model for too many epochs.
5. Lack of regularization.

Step 3: Dropout

- Dropout is a regularization technique used to reduce overfitting. During training, some neurons are randomly deactivated.
- This prevents the network from depending too much on particular neurons and encourages the model to learn more general patterns.

Step 4: Data Augmentation

Data augmentation increases the size and diversity of the training dataset by applying transformations to existing data.

Examples include:

1. Rotation
2. Flipping
3. Cropping
4. Scaling
5. Adding small variations

Data augmentation helps the model learn more general features and improves its performance on unseen data.

Step 5: Importance of Validation Accuracy

Validation accuracy is important because it shows how well the model performs on data that was not used directly for training.

It helps to:

1. Detect overfitting.
2. Measure the generalization ability of the model.
3. Compare different models.
4. Select suitable model parameters.
5. Determine when training should be stopped.

Conclusion:

The model shows overfitting because the training accuracy reaches 97%, while the validation accuracy remains around 72%. The model has learned the training data very well but has poor generalization to unseen data. Techniques such as dropout and data augmentation can be used to reduce overfitting and improve the model's performance.